

Business Sales Performance Analytics

Executive Report with Statistical Analysis

Future Interns

Data Science & Analytics — Task 1
Business Sales Performance Analytics

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1 Executive Summary

This report analyzes **9,994 transactions** across **3 categories** and **4 regions** to identify revenue drivers, profitability patterns, and strategic opportunities for business growth.

1.1 Key Performance Indicators

Metric	Value
Total Sales	\$2,297,201
Total Profit	\$286,397
Profit Margin	12.47%
Total Orders	5,009
Total Customers	793
Average Order Value	\$458.61

Table 1: Business Key Performance Indicators

Key Finding: The business generated **\$2.3M in revenue** with a **12.47% profit margin**, indicating healthy overall operations. However, significant value is being destroyed in specific categories and regions — Furniture (2.49% margin) and the Central region (7.92% margin) require immediate attention.

2 Data Overview

Attribute	Description
Dataset Source	Sample Superstore Dataset
Analysis Period	2014-01-03 to 2017-12-30
Total Transactions	9,994
Unique Products	1,862
Categories	Furniture, Office Supplies, Technology
Regions	West, East, Central, South
Data Quality	0 missing values, 0 duplicates

Table 2: Dataset Characteristics

2.1 Business Questions Addressed

- Q1:** Which products generate the most revenue?
- Q2:** How do sales change over time?
- Q3:** Which categories are most profitable?
- Q4:** Which regions perform best?

Q5: What recommendations can improve business growth?

3 Category Performance

3.1 Revenue and Profit Analysis

Category	Sales	Profit	Sales %	Profit %
Technology	\$836,154	\$145,455	36.4%	50.8%
Furniture	\$742,000	\$18,451	32.3%	6.4%
Office Supplies	\$719,047	\$122,491	31.3%	42.8%

Table 3: Category Performance Summary

Category	Profit Margin	Assessment
Technology	17.40%	Excellent — Market Leader
Office Supplies	17.04%	Strong — Highly Efficient
Furniture	2.49%	Critical — Requires Intervention

Table 4: Category Profit Margin Assessment

3.2 Key Category Insights

Technology — The Star Performer

Generates **36.4% of revenue** but **50.8% of total profit**. Every dollar in Technology sales converts to 17.4 cents in profit. This category deserves increased investment, marketing budget allocation, and inventory prioritization.

Furniture — The Warning Signal

Despite generating **\$742,000 in sales** (32.3% of revenue), Furniture contributes only **\$18,451 in profit** (6.4% of total profit). The 2.49% profit margin indicates severe pricing, discounting, or cost structure issues that require immediate investigation.

4 Sub-Category Deep Dive

4.1 Top Performers by Profit

Category	Sub-Category	Sales	Profit	Margin
Technology	Copiers	\$149,528	\$46,618	31.2%
Office Supplies	Paper	\$78,479	\$34,053	43.4%
Technology	Phones	\$330,007	\$44,515	13.5%
Office Supplies	Binders	\$203,413	\$30,221	14.9%
Technology	Accessories	\$167,380	\$41,937	25.1%

Table 5: Top 5 Most Profitable Sub-Categories

4.2 Loss-Making Sub-Categories

Value Destruction Alert

Several sub-categories are consistently losing money:

- **Tables** (Furniture) — Significant losses despite high sales volume
- **Bookcases** (Furniture) — Negative profit margins
- **Supplies** (Office Supplies) — Low margin performance

These sub-categories require pricing review, cost analysis, or potential discontinuation.

5 Regional Performance

Region	Sales	Profit	Sales %	Profit %	Margin
West	\$725,458	\$108,418	31.6%	37.9%	14.9%
East	\$678,781	\$91,523	29.6%	32.0%	13.5%
Central	\$501,240	\$39,706	21.8%	13.9%	7.9%
South	\$391,722	\$46,749	17.0%	16.3%	11.9%

Table 6: Regional Performance Summary

5.1 Regional Insights

West — The Gold Standard

The West region leads all metrics: highest sales (\$725K), highest profit (\$108K), and best profit margin (14.9%). California alone contributes \$457K in sales with \$76K profit. The West's operational practices should be documented and replicated across other regions.

Central — Critical Underperformance

Despite generating 21.8% of sales, the Central region contributes only 13.9% of profit with a 7.92% margin — nearly half of West's 14.9%. Texas is particularly problematic: \$170K in sales but **-\$25,729 in losses**.

6 Statistical Analysis

6.1 Correlation Analysis

Segment	Correlation (Sales vs Profit)
Overall	0.479
Technology	0.505
Office Supplies	0.551
Furniture	0.174 — Warning
West	0.648
Central	0.641
East	0.524
South	0.007 — Critical

Table 7: Correlation Analysis Results

Statistical Warning

South region shows a near-zero correlation (0.007) between sales and profit — meaning higher sales do NOT translate to higher profits in this region. **Furniture** (0.174) shows the weakest category correlation, confirming that revenue in Furniture does not predict profitability.

6.2 ANOVA — Significance Testing

Test	p-value	Result
Profit ~ Category	3.47×10^{-24}	Highly Significant
Profit ~ Region	0.049	Significant

Table 8: ANOVA Results

Tukey HSD Findings:

- **Technology outperforms Furniture** by approximately \$70 per order ($p < 0.001$)
- **Technology outperforms Office Supplies** by approximately \$58 per order ($p < 0.001$)
- **West outperforms Central** by approximately \$17 per order ($p = 0.043$)
- Other regional differences are not statistically significant

6.3 Linear Regression

Model: Profit = $\beta_0 + \beta_1(\text{Sales}) + \beta_2(\text{Category}) + \beta_3(\text{Region}) + \varepsilon$

Statistic	Simple Model	Multiple Model
R-squared	0.2295	0.2387
Adjusted R-squared	0.2294	0.2383
Sales Coefficient	\$0.18	\$0.18
Intercept	-\$12.73	-\$65.04

Table 9: Regression Model Summary

Interpretation: Every additional dollar in sales generates approximately \$0.18 in profit. Sales alone explains only 23% of profit variation, indicating other factors (discounts, shipping costs, product mix) play significant roles.

6.4 Discount Impact Analysis

Discount Level	Avg Profit/Order	% of Orders
No Discount	+\$66.90	48.0%
Low ($\leq 20\%$)	+\$26.50	38.1%
Medium (21-50%)	-\$110.00	5.4%
High ($>50\%$)	-\$89.40	8.6%

Table 10: Profit by Discount Level

Critical Finding — Discount Correlation: -0.219

14% of all orders have discounts exceeding 20% and lose an average of **\$98 per order**. This represents significant value destruction. The correlation between discount level and profit is **-0.219**, confirming that higher discounts systematically reduce profitability.

7 Recommendations

7.1 Immediate Actions (0–3 Months)

1. **Cap discounts at 20%** — 14% of orders with high discounts are destroying value
2. **Investigate Furniture pricing** — 2.49% margin is unsustainable; review Tables and Book-cases
3. **Audit Texas, Ohio, Pennsylvania** — These states show high sales but significant losses

7.2 Strategic Initiatives (3–12 Months)

4. **Replicate West region best practices** — Document and deploy the 14.9% margin model
5. **Prioritize Technology inventory** — Generates \$70 more profit per order than Furniture
6. **Restructure South region** — Sales-profit correlation near zero requires fundamental changes

7.3 Risk Management

7. **Implement discount approval workflow** — Require management sign-off for discounts above 20%
8. **Exit or restructure loss-making sub-categories** — Discontinue consistently unprofitable lines
9. **Set regional profit margin targets** — Address Central region's 7.92% margin gap

7.4 Estimated Financial Impact

Implementing these recommendations could improve overall profit margin by **2–4 percentage points**, representing approximately **\$46,000–\$92,000 in additional annual profit**.

8 Methodology

Technique	Purpose
Data Cleaning	Missing value detection, duplicate removal, date standardization
Descriptive Statistics	KPI calculation, categorical aggregation
Correlation Analysis	Identify relationships between sales, profit, and discounts
ANOVA with Tukey HSD	Test statistical significance of category and regional differences
Linear Regression	Model profit drivers and quantify relationships
Data Visualization	Professional charts and dashboards

Table 11: Analytical Methodology

Tools Used: R Statistical Computing Environment, tidyverse, ggplot2, scales, broom, knitr

9 Conclusion

The business demonstrates strong overall performance with **\$2.3M in revenue** and **12.47% profit margin**. However, significant value is being destroyed in specific areas that require immediate management attention.

Key Takeaways:

- **Technology and Office Supplies** are the profit engines, each with margins above 17%
- **West region** represents the operational excellence model with 14.9% margins
- **Furniture category** (2.49% margin) and **Central region** (7.92% margin) require strategic intervention
- **Excessive discounting** on 14% of orders systematically erodes profitability
- Statistical evidence confirms that category and regional differences are **statistically significant**

The data clearly indicates that targeted interventions — rather than broad strategic changes — will yield the greatest return on management effort. By focusing on the specific problem areas identified in this analysis, the business can protect and grow profitability while maintaining revenue momentum.

Estimated Impact: \$46,000–\$92,000 in additional annual profit

Report prepared for **Future Interns — Data Science & Analytics Task 1**

Analysis conducted using R Statistical Computing Environment

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